System identification

Given the importance of oscillation frequency for muscle selection and periodicity for muscle power, we evaluate system identification strategies that minimise the error in frequency response characteristics between empirical and analytic.

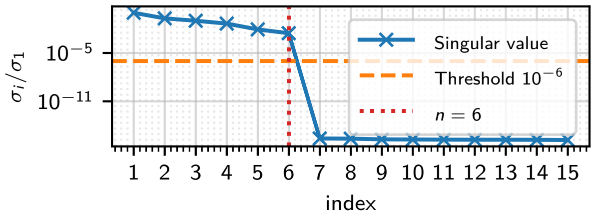

Figure 1: Normalised singular values σ_i/σ_1 of the block Hankel matrix constructed from 55 noise-free Markov parameters (20 block rows, 30 block columns, $p = 2$ inputs, $q = 16$ outputs). The orange dashed line marks the relative threshold 10^{-6}; singular values below this level are indistinguishable from numerical zero under exact noise cancellation. The red dotted line indicates the selected model order $n = 6$. Source

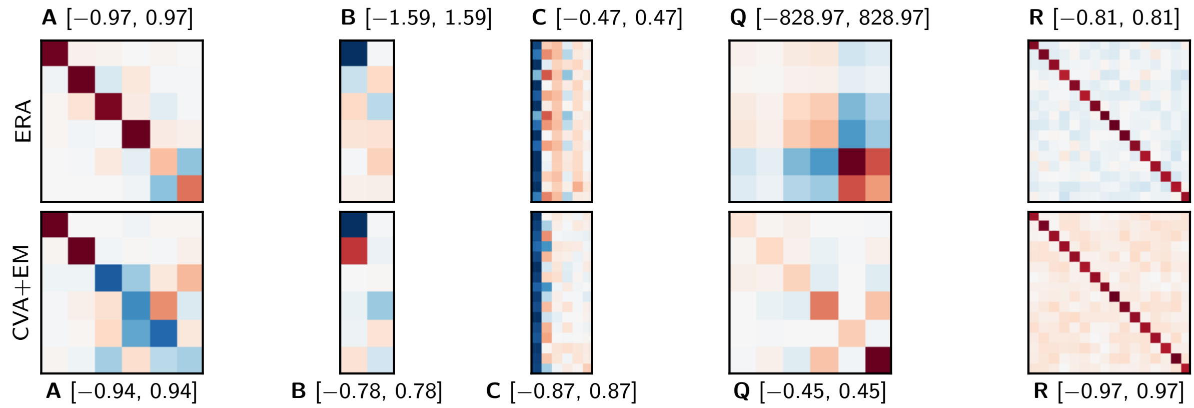

Figure 2: ERA (top row) and CVA+EM (bottom row) estimates of system matrices. Red indicates positive, blue negative; each subplot title shows the symmetric colour range. Source

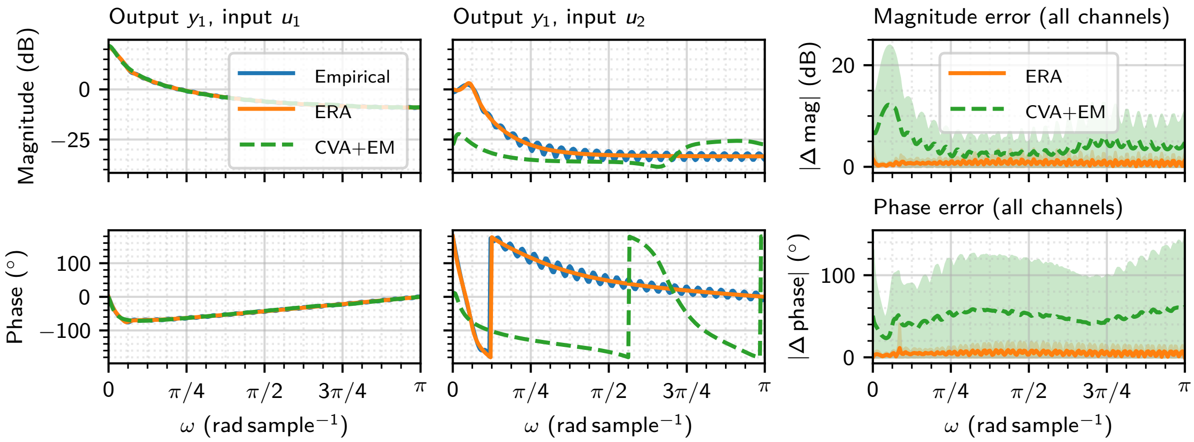

Figure 3: Frequency-response comparison for ERA and CVA+EM identified models. Left and centre columns: Bode magnitude (top) and phase (bottom) for output y_1 driven by inputs u_1 and u_2 respectively, comparing the empirical response (discrete Fourier transform of 55 noise-free Markov parameters, zero-padded to 512 points) against the analytic response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for the ERA and CVA+EM models (order $n = 6$). Right column: absolute magnitude error (top) and absolute phase error (bottom), shown as mean $\pm$ 1 s.d. across all 32 output–input channels for each method. Source

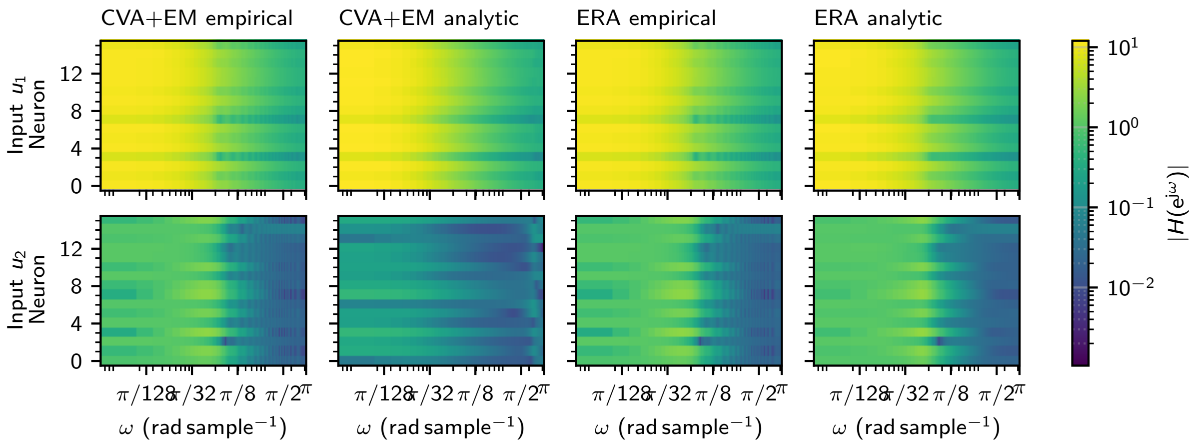

Figure 4: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $q = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample^{-1} (horizontal axis), on a shared logarithmic colour scale. CVA+EM (left 2 columns), ERA (right 2 columns); within each pair, the left column shows the empirical response (DFT of 55 Markov parameters, zero-padded to 512 points) and the right column shows the analytic transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$. Source